

1. Consider the one-period binomial model where a non-dividend-paying stock S has the spot price $S_0 = £18$ and in 1 year time it can either be £27 with probability 0.4 or £9 with probability 0.6. The risk-free discretely compounded interest rate per annum for one year is 5%.

- (a) i. Verify that the market is free of arbitrage. [3 marks]
 ii. Recall that the risk-neutral measure is the probability measure $\mathbb{Q}$, which is determined by probability weights q_u and q_d , such that

$$S_0 = \frac{1}{1+r} \mathbb{E}^{\mathbb{Q}}[S_1]. \quad (1)$$

Using (1) and that $\mathbb{Q}$ is a probability measure, derive the formulae for q_u and q_d . Then find q_u and q_d for this market model. [6 marks]

- (b) Consider a contingent claim Φ that has maturity $T = 1$ year and is written on the stock S . Furthermore, we know the payoff of Φ at maturity is given by

$$\begin{cases} S_T, & \text{if } S_T \leq 10, \\ 20 - S_T, & \text{if } 10 < S_T \leq 20, \\ S_T - 20, & \text{if } S_T > 20. \end{cases}$$

Meanwhile, suppose that we can buy and sell the following European options:

- European Call with strike $K_1 = £20$ and maturity $T = 1$ year.
 - European Put with strike $K_1 = £20$ and maturity $T = 1$ year.
 - European Put with strike $K_2 = £10$ and maturity $T = 1$ year.
- i. Draw a diagram of how the payoff of Φ change according to S_T , the stock price on the maturity date. [4 marks]
 ii. Find a portfolio consisting of three European options listed above that replicates the claim Φ , and justify your answer. You should specify the position and the number of units of each option. (Hint: suppose there are x units Call(T, K_1), y units Put(T, K_1) and z units Put(T, K_2) in this replicating portfolio, what is the payoff?) [9 marks]
 iii. Using the replicating portfolio you found in the previous part and the Law of One Price, find the arbitrage-free price of the contingent claim Φ . Please round your answer to 3 decimal places. [8 marks]

[Total for this question: 30 marks]

Solution

- (a) i. We have $u = 1.5$, $d = 0.5$ and $r = 0.05$. Since

$$d = 0.5 < 1 + r = 1.05 < u = 1.5,$$

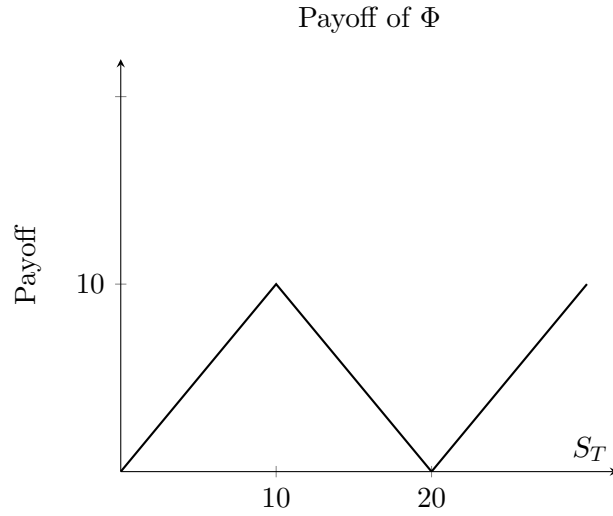
we conclude that the market is arbitrage-free.

- ii. We solve

$$\begin{cases} \frac{1}{1+r}(uS_0q_u + dS_0q_d) = S_0, \\ q_u + q_d = 1, \end{cases}$$

and deduce that

$$q_u = \frac{(1+r) - d}{u - d} = \frac{1.05 - 0.5}{1.5 - 0.5} = 0.55, \quad q_d = 1 - q_u = \frac{u - (1+r)}{u - d} = 0.45.$$



(b) i.

- ii. Suppose that this portfolio contains x units European Call with strike £20, y units European Put with strike £20 and z units European Put with strike £10. Then the payoff of this portfolio at maturity is

$$x(S_T - 20)^+ + y(20 - S_T)^+ + z(10 - S_T)^+ = \begin{cases} y(20 - S_T) + z(10 - S_T), & S_T \leq 10, \\ y(20 - S_T), & 10 < S_T \leq 20, \\ x(S_T - 20), & S_T > 20, \end{cases}$$

which should be equal to the payoff of Φ on each piece, that is we deduce a system of equations

$$\begin{cases} y(20 - S_T) + z(10 - S_T) = S_T, \\ y(20 - S_T) = 20 - S_T, \\ x(S_T - 20) = S_T - 20. \end{cases}$$

This solves $x = y = 1$ and $z = -2$. Therefore, this replicating portfolio consists of

1. one long European Call with strike £20 and maturity $T = 1$ year;
 2. one long European Put with strike £20 and maturity $T = 1$ year;
 3. two short European Put with strike £10 and maturity $T = 1$ year.
- iii. The risk-neutral valuation of each long European Call with strike £20 is

$$\begin{aligned} \Pi_0(Call(20)) &= \frac{1}{1+r} (q_u(27-20)^+ + q_d(9-20)^+) \\ &= \frac{1}{1.05} (0.55 * 7 + 0) \\ &= 3.667. \end{aligned}$$

The risk-neutral valuation of each long European Put with strike £20 is

$$\begin{aligned} \Pi_0(Put(20)) &= \frac{1}{1+r} (q_u(20-27)^+ + q_d(20-9)^+) \\ &= \frac{1}{1.05} (0 + 0.45 * 11) \\ &= 4.714. \end{aligned}$$

The risk-neutral valuation of each long European Put with strike £10 is

$$\begin{aligned}\Pi_0(Put(10)) &= \frac{1}{1+r} (q_u(10 - 27)^+ + q_d(10 - 9)^+) \\ &= \frac{1}{1.05} (0 + 0.45 * 1) \\ &= 0.429.\end{aligned}$$

Therefore, by the previous part, we conclude that $\Pi_0(\Phi) = 3.667 + 4.714 - 2 * 0.429 = 7.523$. So the arbitrage-free price for the claim Φ is £7.523.

DRAFT

2. The price of the stock S today is £100, and it is expected to increase by 20% or decrease by 10% every 3 months over the next half year. The continuously compounded risk-free interest rate per annum is 4%.

- (a) i. Show that the market is free of arbitrage. [3 marks]
 ii. Recall that under the continuously compounded interest rate, the risk-neutral measure $\mathbb{Q}$ consists of weights q_u and q_d such that

$$S_0 = e^{-r\Delta t} \mathbb{E}^{\mathbb{Q}}[S_1].$$

Find the risk-neutral probabilities q_u and q_d for this two-period market. [4 marks]

- (b) Find the arbitrage-free price of a 6-month European call option Φ with strike price $K = £100$.
 (c) Find the arbitrage-free price of a 6-month European-type put option Ψ with strike price $K = £100$, whose payoff function is

$$g(S_T) = ((K - S_T)^+)^2.$$

[7 marks]

- (d) Find the arbitrage-free price of a 6-month American put option X with strike price $K = £110$. Is there any early exercise opportunity?

[10 marks]

[Total for this question: 31 marks]

Solution

- (a) i. We have $d = 0.9$, $u = 1.2$, $r = 0.04$ and $\Delta t = \frac{1}{4}$. Then since

$$d = 0.9 < e^{r\Delta t} = 1.01 < u = 1.2,$$

the market is free of arbitrage.

ii.

$$q_u = \frac{e^{r\Delta t} - d}{u - d} = \frac{11}{30}, \quad q_d = 1 - q_u = \frac{19}{30}.$$

- (b) Using the payoff function, we have

$$f(S_2(uu)) = (144 - 100)^+ = 44,$$

$$f(S_2(ud)) = (108 - 100)^+ = 8,$$

$$f(S_2(dd)) = (81 - 100)^+ = 0.$$

Therefore, the risk-neutral valuation is

$$\begin{aligned} \Pi_0(\Phi) &= e^{-2r\Delta t} \mathbb{E}^{\mathbb{Q}}[f(S_2)] \\ &= e^{-2r\Delta t} (q_u^2 f(S_2(uu)) + 2q_u q_d f(S_2(ud)) + q_d^2 f(S_2(dd))) \\ &= e^{-0.02} \left(\left(\frac{11}{30} \right)^2 * 44 + 2 \frac{11}{30} \frac{19}{30} * 8 + 0 \right) \\ &= 9.452. \end{aligned}$$

Therefore, the arbitrage-free price of Φ is £9.452.

(c) We have

$$\begin{aligned} g(S_2(uu)) &= ((100 - 144)^+)^2 = 0, \\ g(S_2(ud)) &= ((100 - 108)^+)^2 = 0, \\ g(S_2(dd)) &= ((100 - 81)^+)^2 = 361. \end{aligned}$$

Therefore,

$$\begin{aligned} \Pi_0(\Psi) &= e^{-2r\Delta t} \mathbb{E}^{\mathbb{Q}}[g(S_2)] \\ &= e^{-2r\Delta t} (q_u^2 g(S_2(uu)) + 2q_u q_d g(S_2(ud)) + q_d^2 g(S_2(dd))) \\ &= e^{-0.02} \left(0 + 0 + \left(\frac{19}{30}\right)^2 * 361 \right) \\ &= 141.541. \end{aligned}$$

Therefore, the arbitrage-free price of Ψ is £141.541.

(d) We first notice that

$$\begin{aligned} h(S_2(uu)) &= (110 - 144)^+ = 0, \\ h(S_2(ud)) &= (110 - 108)^+ = 2, \\ h(S_2(dd)) &= (110 - 81)^+ = 29. \end{aligned}$$

At $S_1(u)$:

- Risk-neutral valuation:

$$e^{-r\Delta t} (q_u h(S_2(uu)) + q_d h(S_2(ud))) = e^{-0.01} \left(\frac{11}{30} * 0 + \frac{19}{30} * 2 \right) = 1.253.$$

- Intrinsic value:

$$h(S_1(u)) = (110 - 120)^+ = 0.$$

- $\Pi_1(X)|_{S_1(u)} = \max\{1.253, 0\} = 1.253$, so no early exercise.

At $S_1(d)$:

- Risk-neutral valuation:

$$e^{-r\Delta t} (q_u h(S_2(ud)) + q_d h(S_2(dd))) = e^{-0.01} \left(\frac{11}{30} * 2 + \frac{19}{30} * 29 \right) = 18.9.$$

- Intrinsic value:

$$h(S_1(d)) = (110 - 90)^+ = 20.$$

- $\Pi_1(X)|_{S_1(d)} = \max\{18.9, 20\} = 20$, so there is an early exercise opportunity.

At S_0 :

- Risk-neutral valuation:

$$e^{-r\Delta t} (q_u \Pi_1(X)|_{S_1(u)} + q_d \Pi_1(X)|_{S_1(d)}) = e^{-0.01} \left(\frac{11}{30} * 1.253 + \frac{19}{30} * 20 \right) = 12.989.$$

- Intrinsic value:

$$h(S_0) = (110 - 100)^+ = 10.$$

- $\Pi_0(X) = \max\{12.989, 10\} = 12.989$, so no early exercise.

In conclusion, we have that the price of X is £12.989 and there exists an early exercise opportunity.

3. Let $(B_t)_{t \geq 0}$ be a Brownian motion on some probability space.

(a) Fix some $\alpha > -\frac{1}{2}$. Define for each $t \geq 0$,

$$X_t := \int_0^t s^\alpha dB_s.$$

Find $\mathbb{E}[X_t]$ and $\text{Var}(X_t)$.

[7 marks]

(b) Show that the stochastic process $(X_t)_{t \geq 0}$ is a Brownian motion if and only if $\alpha = 0$. [4 marks]

[Total for this question: 11 marks]

Solution

(a) By the definition of X , we see that

$$\mathbb{E}[X_t] = 0.$$

Meanwhile, by Itô isometry, we have

$$\begin{aligned} \text{Var}(X_t) &= \mathbb{E}[X_t^2] \\ &= \mathbb{E} \left[\left(\int_0^t s^\alpha dB_s \right)^2 \right] \\ &= \int_0^t s^{2\alpha} ds \\ &= \frac{1}{2\alpha + 1} t^{2\alpha+1}. \end{aligned}$$

(b) When $\alpha = 0$, it follows that

$$X_t = \int_0^t 1 dB_s = B_t,$$

so X is a Brownian motion. When $\alpha \neq 0$, we see from the previous part that

$$\frac{1}{2\alpha + 1} t^{2\alpha+1} \neq t,$$

and thus X cannot be a Brownian motion.

4. Suppose $T, r, \sigma > 0$ are constants. Let $(W_t)_{t \geq 0}$ be a Brownian motion. Consider the following stochastic differential equation:

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad (2)$$

and $S_0 > 0$ is a positive real number.

- (a) By applying Itô's formula to $Y_t = \ln S_t$, prove that the solution of (2) is

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right). \quad (3)$$

[7 marks]

- (b) Using (3), show that for fixed $K > 0$,

$$\mathbb{P}(S_T > K) = N(d_-), \quad (4)$$

where $N(x)$ is the cumulative distribution function (CDF) for the standard normal distribution $N(0, 1)$ and

$$d_- = \frac{\ln \left(\frac{S_0}{K} \right) + \left(r - \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}}.$$

[7 marks]

- (c) Using (3), show that for every $K > 0$,

$$\mathbb{E}[S_T 1_{\{S_T > K\}}] = S_0 e^{rT} N(d_+), \quad (5)$$

where $N(x)$ is the cumulative distribution function (CDF) for the standard normal distribution $N(0, 1)$ and

$$d_+ = d_- + \sigma \sqrt{T} = \frac{\ln \left(\frac{S_0}{K} \right) + \left(r + \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}}.$$

[9 marks]

- (d) Suppose that the stock price at time t is given by the stochastic process S_t in (3), and $S_0 > 0$ is the current price of the stock. The bank account pays interest continuously in time at a risk-free interest rate $r > 0$.

Using (4) and (5), compute the Black-Scholes price (or arbitrage-free price) $\Pi_0(\text{Call}(T, K))$ of the European Call option with strike K , maturity T , and payoff $\Phi(S_T) = (S_T - K)^+$. [5 marks]

[Total for this question: 28 marks]

Solution

- (a) Let us consider $Y_t = \ln S_t$, i.e. $f(t, x) = \ln x$ in the statement of Itô's formula. Computing the derivatives appearing in the Itô formula

$$\partial_t f = 0, \quad \partial_x f = \frac{1}{x}, \quad \partial_{xx} f = -\frac{1}{x^2}.$$

Now we have

$$dY_t = \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \langle dS_t, dS_t \rangle$$

One can easily compute $\langle dS_t, dS_t \rangle = \sigma^2 S_t^2 dt$, so that using the SDE for S , we have

$$\begin{aligned} dY_t &= \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \sigma^2 S_t^2 dt \\ &= \frac{1}{S_t} (r S_t dt + \sigma S_t dW_t) - \frac{\sigma^2}{2} dt, \end{aligned}$$

which leads to the following SDE:

$$\begin{aligned} dY_t &= \left(r - \frac{\sigma^2}{2} \right) dt + \sigma dW_t, \\ Y_0 &= \ln S_0, \end{aligned}$$

that is,

$$Y_t = \ln S_0 + \left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t.$$

Finally, by taking exponential, we see that

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right).$$

(b) For any $K > 0$,

$$\begin{aligned} \mathbb{P}(S_T > K) &= \mathbb{P} \left(S_0 \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T \right\} > K \right) \\ &= \mathbb{P} \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T > \ln \left(\frac{K}{S_0} \right) \right) \\ &= \mathbb{P} \left(W_T > \frac{\ln \left(\frac{K}{S_0} \right) - \left(r - \frac{\sigma^2}{2} \right) T}{\sigma} \right) \\ &= \mathbb{P} \left(\sqrt{T} N(0, 1) > \frac{\ln \left(\frac{K}{S_0} \right) - \left(r - \frac{\sigma^2}{2} \right) T}{\sigma} \right) \\ &= \mathbb{P}(N(0, 1) > -d_-) \\ &= N(d_-). \end{aligned}$$

(c) Using (3), we have

$$\begin{aligned} \mathbb{E}(S_T 1_{\{S_T > K\}}) &= \mathbb{E} \left(S_0 \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T \right\} 1_{\{S_T > K\}} \right) \\ &= \mathbb{E} \left(S_0 \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} N(0, 1) \right\} 1_{\{N(0, 1) > -d_-\}} \right) \\ &= S_0 \int_{-d_-}^{\infty} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} x \right\} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= S_0 e^{rT} \int_{-d_-}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x - \sigma \sqrt{T})^2}{2}} dx \\ &= S_0 e^{rT} \int_{-d_- - \sigma \sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= S_0 e^{rT} \mathbb{P}(N(0, 1) > -d_+) \\ &= S_0 e^{rT} N(d_+). \end{aligned}$$

(d) We see that

$$\begin{aligned}
 \Pi_0(\text{Call}(T, K)) &= e^{-rT} \mathbb{E}(\Phi(S_T)) \\
 &= e^{-rT} \mathbb{E}((S_T - K)^+) \\
 &= e^{-rT} \mathbb{E}((S_T - K)1_{\{S_T > K\}}) \\
 &= e^{-rT} \mathbb{E}(S_T 1_{\{S_T > K\}}) - K e^{-rT} \mathbb{P}(S_T > K).
 \end{aligned} \tag{6}$$

Now substitute (4) and (5) in (6), we deduce that

$$\Pi_0(\text{Call}(T, K)) = S_0 N(d_+) - K e^{-rT} N(d_-).$$

DRAFT